

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
5	(a)		Net / resultant / [vector] sum of / total force (1) Correct example showing net of two or more forces numerically, algebraically or by convincing diagram (1)	1	1		2		
	(b)	(i)	Weight of block = 14 715 [N] (1) $a = \frac{(16\,000 - 14\,715)}{1500}$ (1) substitution and re-arrangement $a = 0.86 \text{ [m s}^{-2}\text{]} (1)$ Alternative: $a = \frac{16\,000}{1500} = 10.67 \text{ [m s}^{-2}\text{]} (1)$ $a = 10.67 - 9.81 (1)$ $a = 0.86 \text{ [m s}^{-2}\text{]} (1)$	1	1 1		3	3	
		(ii)	Depends on acceleration (1) may exceed 16 000 N, so no (1)			2	2		
	(c)		Stress = $2 \times 10^{11} \times 3.2 \times 10^{-5} = 6.4 \times 10^6 \text{ [N m}^{-2}\text{]} (1)$ Tension = $6.4 \times 10^6 \times 2 \times 10^{-3} = 12\,800 \text{ [N]} (1)$ Award 2 marks for determining T directly using strain $\times E \times A$ Decelerating – value not required but may be seen (1) ecf from T Because $T < 14\,715 \text{ [N]}$ (can be implied) (1) Alternative for 3rd and 4th marks $\Sigma F = 12\,800 - 14\,700 = -1\,900 \text{ [N]} (1)$ Negative value indicates a negative acceleration so the load must be decelerating (1)		4		4	2	